

6.3 Automating Things

2. eventloop.py

What the script does: In order:

- catch input errors
- open given file
- load ROOT tree
- make new `.root` file to save to later
- make a new histogram object with `ROOT.TH1D(...)`
- loop over all collisions:
 - in each, read out all data for the collision (check for success)
 - fill `vxp_z` for that collision into the histogram object
- write all histograms inside the new `.root` file
- by default, create a popup window/canvas with the created histogram
 - wait for user input before the popup is closed

We limited to 100,000 entries.

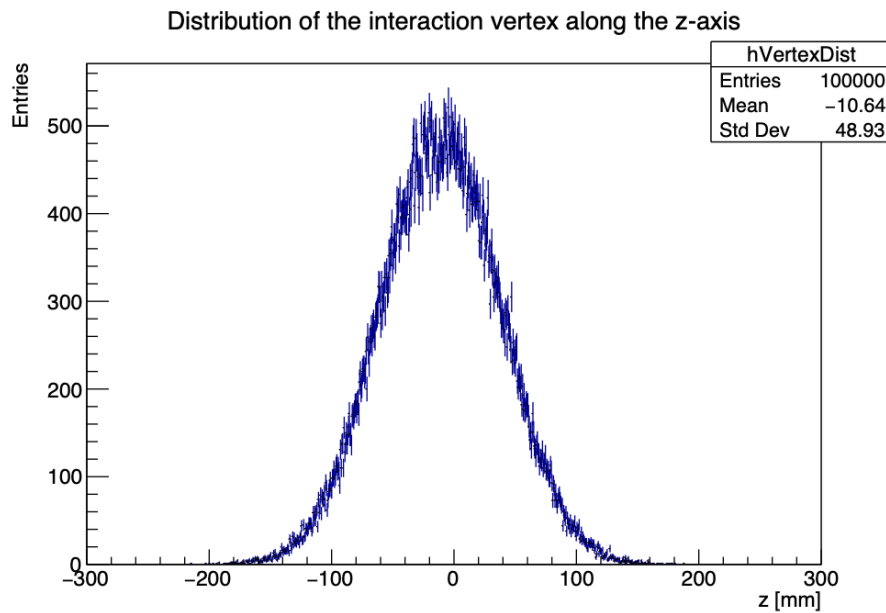


Figure 1: `vxp_z` graph

Converted MeV to GeV in below graph:

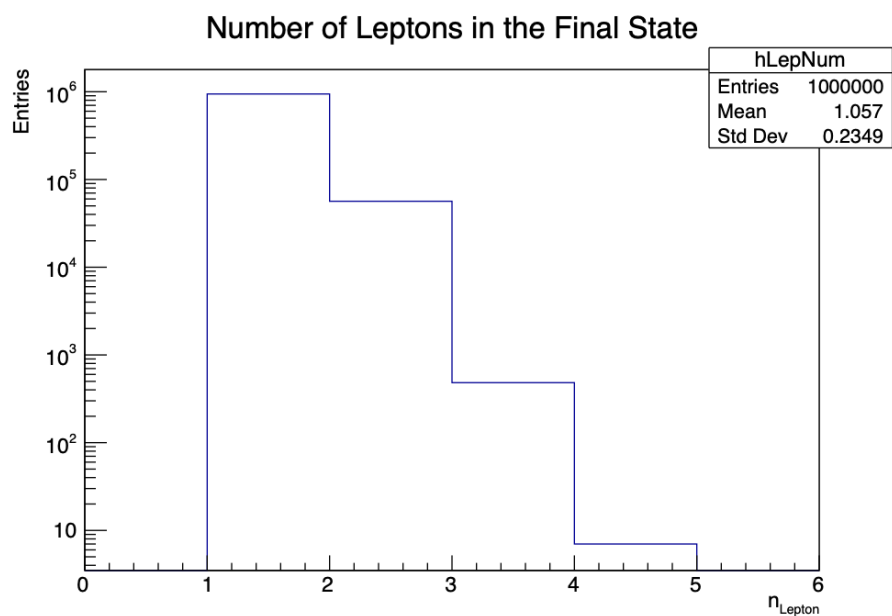


Figure 2: lep_n graph

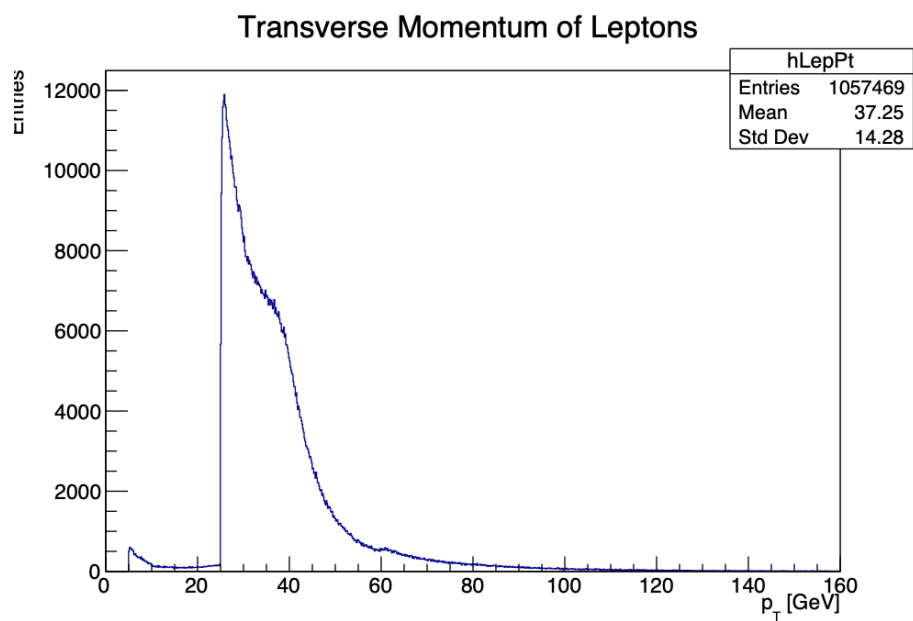


Figure 3: lep_pt graph

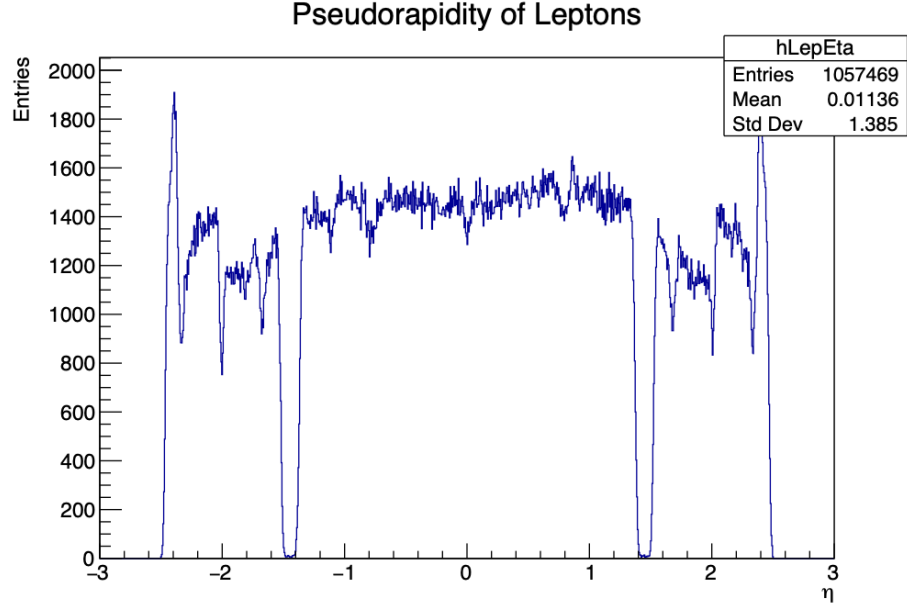


Figure 4: lep_eta graph

3. Invariant Mass

Rapidity:

$$y = \frac{1}{2} \ln\left(\frac{E + p_z}{E - p_z}\right)$$

By trigonometry:

$$p_z = |\vec{p}| \cdot \cos(\theta)$$

For low ratio of jet mass / jet energy (highly relativistic particles):

$$E \approx p$$

$$y \approx \frac{1}{2} \ln\left(\frac{p + p_z}{p - p_z}\right)$$

$$y \approx \frac{1}{2} \ln\left(\frac{p + p \cdot \cos(\theta)}{p - p \cdot \cos(\theta)}\right)$$

$$y \approx \frac{1}{2} \ln\left(\frac{1 + \cos(\theta)}{1 - \cos(\theta)}\right)$$

$$y \approx \frac{1}{2} \ln\left(\frac{2 \cos^2(\frac{\theta}{2})}{2 \sin^2(\frac{\theta}{2})}\right)$$

$$y \approx \frac{1}{2} \ln(\tan^{-2}(\frac{\theta}{2}))$$

This gives us the pseudorapidity:

$$\eta = -\ln\left(\tan\frac{\theta}{2}\right) \approx y$$

Invariant mass of the two leading leptons:

$$m_0^2 c^2 = \frac{E^2}{c^2} - \vec{p}^2$$

$$p_z = p_T \cdot \cot(\theta)$$

$$p_z = p_T \cdot \frac{\cos(\theta)}{\sin(\theta)}$$

$$p_z = p_T \cdot \frac{\cos^2(\frac{\theta}{2}) - \sin^2(\frac{\theta}{2})}{2 \sin(\frac{\theta}{2}) \cos(\frac{\theta}{2})}$$

$$p_z = p_T \cdot \frac{\cot(\frac{\theta}{2}) - \tan(\frac{\theta}{2})}{2}$$

$$p_z = p_T \cdot \frac{e^\eta - e^{-\eta}}{2}$$

$$p_z = p_T \cdot \sinh(\eta)$$

Then:

$$p_T^2 = p_x^2 + p_y^2 = p_T^2 + p_z^2 = p_T^2 \cdot (1 + \sinh^2(\eta))$$

Now substitute into the equation for invariant mass:

$$m_0^2 = \frac{E^2}{c^4} - \frac{\vec{p}^2}{c^2} m_0^2 = \frac{E^2}{c^4} - p_T^2 \cdot (1 + \sinh^2(\eta)) / c^2$$

We apply the equation to variables and graph histogram of result. We restrict/disallow negative m_0^2 so that all masses are positive.

4. TLorentzVector

Both methods do the same thing and produce very similar histograms.

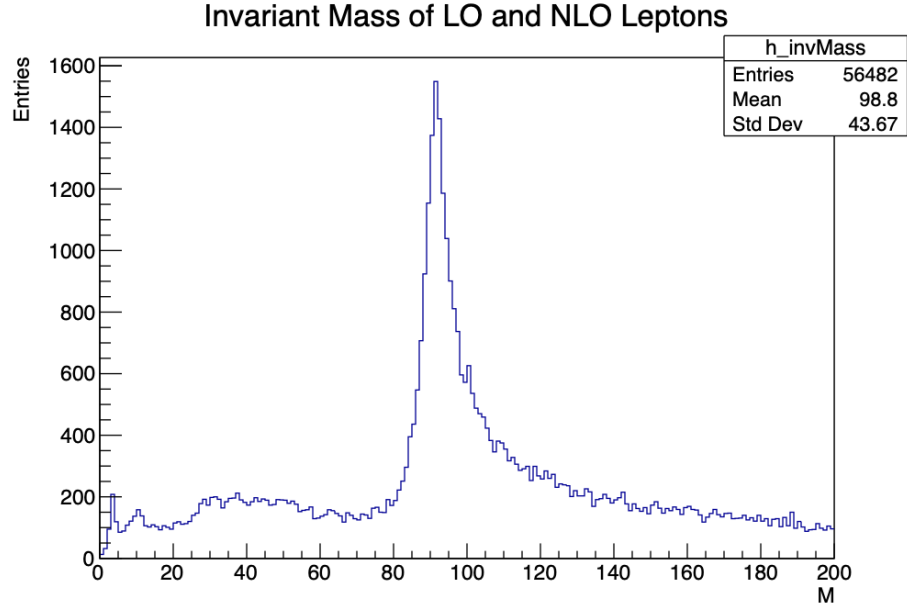


Figure 5: Invariant Mass graph

5. Expected Distribution

A priori, one might expect a symmetric distribution around the expected value of ~ 90 GeV. For example, a Gaussian or Breit-Wigner distribution.

This is generally true, but we can also see smaller peaks at around ~ 3 GeV and ~ 9.5 GeV. These coincide with the rest masses of the J/Ψ and Υ mesons. These can decay into a γ^* , which can then become two leptons (meaning the event is included in the data, even after filtering).

We also see a “hill” due to the 25 GeV cutoff.

MC simulation to find what we would expect from the data. (Simulated distribution matches quite well to measured distribution.)

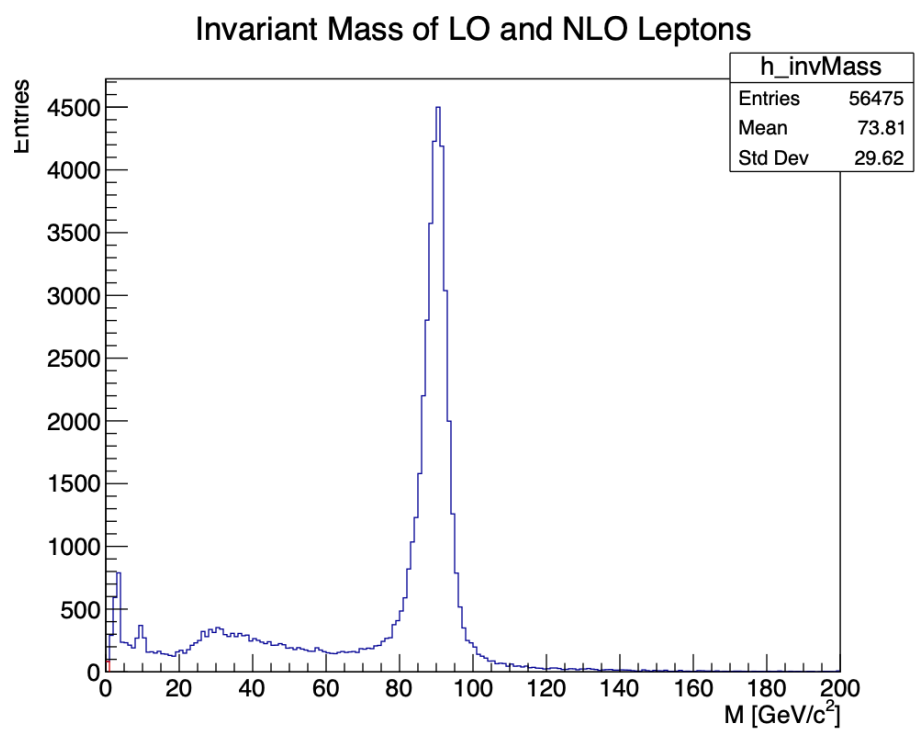


Figure 6: Invariant Mass graph

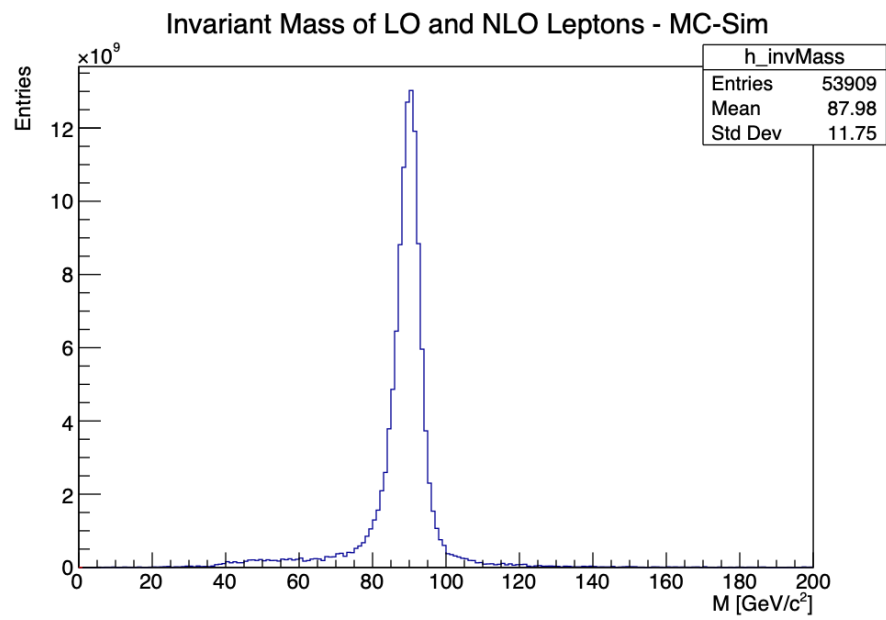


Figure 7: Invariant Mass graph